DSA INVESTIGATION ASSIGNMENT REPORT

The detective was dispatched onto the scene with magnifying glass in hand, lab equipment in the bag. Initial DNA scrubbing returned no answers and so the security footage and logs were analyzed. The computer lacked a sufficient application for the search he required hence he got to programming. Detective Dennis tried implementing a sorting algorithm for the large amount of data that the log provided. He made an app called Eyesharp which allowed him to sort through any field. The admin was impressed by his tech savvyness and considered offering the job to him. Declining the offer he continued the investigation. With the now sorted algorithm he could use his implementation of his generic binary search he had created. He could search on any of the fields that the log provided, the important ones were the Last Entered, and Last Exit into the engine room. Creating the generic binary search was simple enough but creating the more advanced search he had hoped was tough. He tried to make the search be able to find the closest values that he entered. The difficulty was getting the hashmap to work correctly and would give wrong values, rounding errors he surmised. After two days he settled for the first binary search he created which was appropriate however it required him to find two time ranges to search for on both entrance and exit time which took quite some time, but was eased by the search function that was installed on his editor. With the four values in hand he could search on both fields. The search provided him with a couple of suspects within the time frame given to him. A list of 8 suspects, some entered during the timeframe while some left during that time, only one person entered and left during that timeframe. Alfred Pugsley. The detective wanted a visually pleasing way to display all the employees and so created a tree and hashmap to aid him. He decided to go with the Tree since it looked the most presentable. A Depths first search and Breadth first search tree was created and compared in times of computing time. It seemed to be about the same time but slower than the binary search. He made the trees take in DateTime instead of being generic since it suited his end goal but he could just as well made it generic with some extra code. It seemed though Unity's editor provided highly randomized returns in computing times for the algorithms since a lot of cleanup is done behind the scenes. But the detective tried his best to get some values. These are the average numbers but are a bit inflated by unity lag, so to get better values a more isolated test is required.

Search	Average(Nanoseconds)
Binary	7034
DFS	200k-300k
BFS	200k-300k

Suspect analysis
Group 1
Jasper Wellington ID:47 Entrance Time 2234-08-09 22:13:17 Last Exit 2234-08-04T07:14:06
Nem Wellington ID:288 Entrance Time 2234-08-09 22:13:17 Last Exit 2234-08-02T09:16:49
Helena Blackwood ID:764 Entrance Time 2234-08-09 22:13:17 Last Exit 2234-08-03T09:43:27
These three entered at the appropriate time however didn't leave for a while until the logs were paused and questioned by the authority. They have an alibi and they all were cleared of suspicion.

Group 2
Helena Wickham ID:621 Entrance Time 2234-08-09T23:15:58 Last Exit 2234-08-03T12:42:23
Entered later, did not commit the crime but was questioned for the possibility of trying to remove evidence

Group 3
Arthur Holloway ID:118 Entrance Time 2234-08-02T06:11:28 Last Exit 2234-08-09T22:14:54
Beatrice Graves ID:514 Entrance Time 2234-08-03T12:51:23 Last Exit 2234-08-09T22:14:54
Alfred Underwood ID:792 Entrance Time 2234-08-08T21:03:38 Last Exit 2234-08-09T22:14:54

These three could have helped with the crime or been witnesses so they were thoroughly questioned but gave no answer, however they gave no leeway into the investigation but they reported seeing Alfred Pugsley amongst them.

Alfred Pugsley was questioned and his time correlated with the suspected timeframe of the murder and he admitted to the crime.